A Two-Stage Unified Refinement Framework With RS-Specific Priors for Remote Sensing Open-Vocabulary Segmentation

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***Abstract*—Vision–language pretrained models, particularly CLIP, have demonstrated remarkable zero-shot transfer capabilities across various image-level tasks, catalyzing the advancement of open-vocabulary semantic segmentation (OVSS) in remote sensing (RS). However, the direct deployment of CLIP to the RS domain is inherently constrained by the profound domain shift between terrestrial and overhead perspectives, as well as the intricate geometric heterogeneities regarding scale and orientation. To circumvent these limitations, we propose CDSeg, a robust framework tailored for RSOVSS. Central to this architecture is the dual-domain feature compensation module (DDFCM), which integrates DINOv3 weights, pretrained on large-scale RS benchmarks, to augment CLIP with domain-specific semantic priors, effectively bridging the natural-to-satellite knowledge gap. Furthermore, we introduce a MambaVision-driven cross-feature fine-grained interaction module (CFFIM) to facilitate a unified refinement of spatial and category attributes, leveraging long-range dependency modeling to enhance the model's discriminative power in unseen environments. To robustly manage the complexities of diverse orientations and scales, CDSeg incorporates a direction-aware rotation strategy and a wavelet-cross-attention-enhanced module (WCAEM) for high-fidelity multiscale feature decoding. Empirical evaluations on four public benchmarks demonstrate that CDSeg achieves state-of-the-art (SOTA) performance, while extensive ablation studies substantiate the synergistic contribution and indispensability of each component.**

***Index Terms*—CDSeg, cross-feature fine-grained interaction module (CFFIM), dual-domain feature compensation module (DDFCM), remote sensing (RS) open-vocabulary segmentation, wavelet-cross-attention-enhanced module (WCAEM).**

I. Introduction

RELATIVE to the unimodal remote sensing (RS) semantic segmentation [1], [2], [3], [4], multimodal RS semantic segmentation [5], [6], [7], [8], [9] (opt-x image modal) and referring RS semantic segmentation [10], [11], [12], [13] (segmenting objects related to textual descriptions) involve fully supervised training and testing on datasets with fixed, identical categories shown in Fig. 1(a) and (b). The RS open-vocabulary semantic segmentation (OVSS) [14], [15], [16], [17] task illustrated in Fig. 1(a) and (c) not only can identify all pixels in the image that match the text definitions but also overcomes the limitations of predefined closed-set labels. This is of great significance for building autonomous, interactive, and zero-shot generalizable geographic information intelligence (Geo-AI).

The widespread rise of vision–language models has greatly improved the accuracy of OVSS models in recognizing category information that did not appear in the training set. Therefore, OVSS has currently achieved great success in natural scenes, and many two-stage methods [18], [19], [20] have emerged. Ge et al. [21] introduced CRTNet, comprising a region-to-text decoder and mask language distillation, to address the reliance on predefined category sets and the constraints of limited vocabularies in open-vocabulary segmentation. To tackle the overlooked diversity of multihead self-attention and the issue of feature redundancy in training-free open-vocabulary segmentation, Duan et al. [22] developed DIH-CLIP, which incorporates selective single-head attention enhancement. However, the above methods are all designed for natural scenes and cannot be well adapted directly to RS scenarios. Fortunately, an increasing number of researchers have begun exploring the application and adaptation of OVSS in the field of RS, achieving promising foundational results. For example, Li et al. [23] proposed the training-free SegEarth-OV method, which includes innovations such as feature recalibration, spatial structure enhancement, and cross-modal alignment optimization. Li et al. [24] introduced the SegEarth-OV3 method, featuring innovations like a mask fusion strategy

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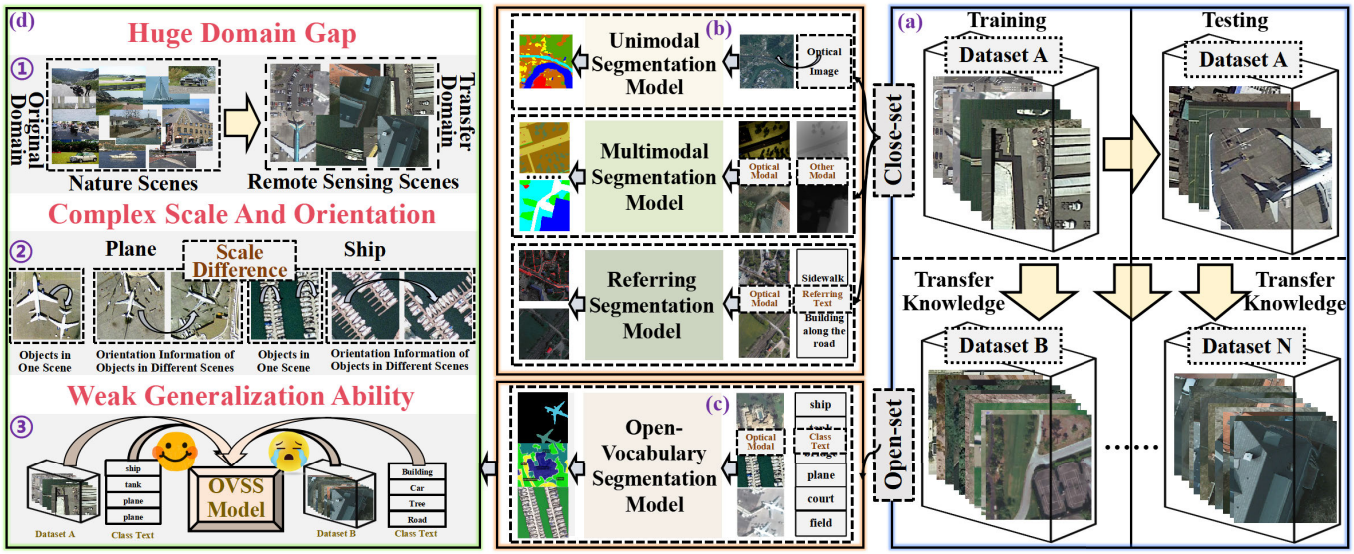


Fig. 1. This image provides a detailed explanation of the differences between open-set and closed-set tasks (a)–(c) and summarizes the three main challenges currently faced by RSOVSS (d). The order of the image is from right to left.

and an existence score filtering mechanism. These works have laid a solid foundation for our research and validated the feasibility of RS; however, none of them have addressed the following inherent shortcomings present in RSOVSS.

1) *Huge Domain Gap:* As illustrated in Fig. 1(d), existing visual language models (VLMs) are primarily pretrained on natural scene data from the Internet, and their representation space fundamentally differs from that of the RS observation domain. This gap is reflected in the image dimension by the drastic viewpoint changes between natural scenes and RS scenes. In the semantic dimension, although the same object (e.g., car) shares text labels in both domains, its visual representations (side features versus top contours) differ significantly. This inconsistency in cross-domain features leads to a severe semantic alignment challenge when transferring pretrained knowledge to the complex semantic environment of RS.
2) *Complex Scale and Orientation Information:* As illustrated in Fig. 1(d), RS objects exhibit strong spatial geometric heterogeneity. On one hand, the targets display arbitrary orientation; not only do similar objects in different scenes have vastly different directions, but targets within the same scene also show significant uneven distribution of rotation. On the other hand, due to the strong influence of sensor altitude and resolution fluctuations, there is severe scale variation in the pixel spans of the same semantic category. Existing models often cannot model rotation invariance and aggregate multiscale features, making it difficult to maintain robust feature responses under complex geometric transformations.
3) *Weak Generalization Ability:* As illustrated in Fig. 1(d), the current model often struggles to establish precise semantic associations between entirely new text embeddings and highly complex RS land cover backgrounds without prior guidance; it faces significant semantic confusion when handling out-of-distribution (OOD) data.

To resolve the aforementioned challenges, we propose a novel OVSS framework for RS, termed CDSeg, relying upon two robust feature extraction architectures: DINOv3 [25] and CLIP [26]. First, CDSeg introduces a dual-domain feature compensation module (DDFCM). DDFCM leverages the pretrained DINOv3 backbone, which is abundant with RS priors, and, inspired by the LoRA [27] parameter fine-tuning approach, embeds RS domain-specific knowledge into the frozen CLIP feature extraction architecture within the proposed model. In addition, we devise a multidirectional, multiscale feature fusion and decoding strategy and formulate the wavelet-cross-attention-enhanced module (WCAEM). Through cross-attention and frequency-domain hierarchical feature decomposition, WCAEM substantially amplifies the representation of directional and scale-related features across RS scenes. Finally, to bolster the model's generalization capacity concerning unknown scenes, we introduce the cross-feature fine-grained interaction module (CFFIM), which establishes a unified interaction strategy for both spatial and class refinement. Via a state transition matrix, CFFIM synchronously reinforces spatial coherence alongside textual semantic coherence at both pixel and text semantic levels. This unified processing paradigm enforces structural similarity among correlated features, effectively driving the two-stage multimodal aggregation and guaranteeing that during zero-shot transfer, the model can precisely recognize target objects within unseen scenes. In summary, the contributions of this work are as follows.

1) To adapt to the RSOVSS task, we propose a novel RSOVSS framework termed CDSeg. Through extensive comparative and ablation experiments, we demonstrate the effectiveness and indispensability of CDSeg.
2) Without diminishing the inherent generalized comprehension capabilities of CLIP, to bridge the visual representation gap between natural and RS domains, we

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introduce a DDFCM built upon the DINOv3 backbone network pretrained across RS scenes, accurately integrating domain-specific knowledge from the RS field into the visual features.

3) To mitigate the suboptimal generalization capacity of RSOVSS, CDSeg introduces a novel CFFIM that elevates the model's comprehension regarding unknown scenes across 2-D: spatial structural relationships alongside categorical text semantic interactions.
4) Finally, CDSeg utilizes a novel scene rotation strategy and a cross-scale decoding module to capture and interpret scale and directional information inherent within RS scenes.

This article is structured as follows. Section II provides an overview of the related works. Section III introduces the proposed CDSeg framework and its associated components. The experimental results and discussion analysis are presented in Section IV. Lastly, the conclusion is drawn in Section V.

II. Related Works

A. OVSS in Natural Scenes

Open vocabulary tasks initially developed in the natural scene domain. Although related research cannot be directly applied to the RS field, it still lays the foundation for the development of open vocabulary semantic segmentation (OVSS) in RS. Xie et al. [28] proposed the SED method, which includes hierarchical encoded cost map generation, a progressive fusion decoder, and an early category rejection mechanism, to address the challenges of ViT backbone networks lacking local spatial information and having high inference computational costs in open vocabulary semantic segmentation. Wang et al. [29] introduced the USE method, featuring an automated data pipeline for generating fine-grained segmentation–text pairs and a lightweight universal segmentation embedding model, to tackle the difficulty of accurately classifying image segmentation blocks of varying granularity in open vocabulary segmentation. Sun et al. [30] proposed the CLIPer method, which incorporates a shallow feature fusion mechanism and a refinement compensation module, to overcome the issues of the CLIP model lacking spatial structural information and producing coarse segmentation edges in open vocabulary semantic segmentation. Shan et al. [31] developed the EBSeg method, introducing an adaptive balanced decoder and a semantic structure consistency loss, to solve the problems of models overfitting training classes and limited generalization to new classes in open vocabulary semantic segmentation. Park et al. [32] proposed a personalized segmentation approach that includes personal concept parsing, context prompt generation, and an adaptive fusion module to address the difficulty of recognizing specific personal concepts in open vocabulary semantic segmentation. Shin et al. [33] introduced PixelCLIP to mitigate the heavy reliance on semantic labels and the inherent difficulty in precise object localization within OVSS. Their approach incorporates an online clustering algorithm, learnable class names, and unsupervised mask guidance. To address the prohibitive training costs of auxiliary segmentation networks and the challenges of pixel-level feature extraction, Li et al. [34] developed RPN, which features two core innovations: a relation prompting module and a relation prompting network. Furthermore, observing that most methods rely solely on vision–language pretraining while neglecting the rich priors of large models, Shi et al. [35] proposed LLMFormer, integrating semantic, scaled-visual, and relational attention mechanisms.

B. OVSS in RS

Dutta et al. [36] proposed the AerOSeg method, which includes multirotational image correlation features, spatial and category refinement modules, and SAM-generated automation, to address the challenges of varying object scales and complex orientations in open-vocabulary segmentation of RS images. Cao et al. [37] introduced the OVRS method, featuring a rotation-aggregated similarity calculation module and a multiscale feature integration strategy, to tackle difficulties caused by rapidly changing object directions and significant scale differences in open-vocabulary segmentation of RS images. Huang et al. [15] developed the KG-OVRSeg method, incorporating knowledge graph structural enhancement, category-aware attention, and a novel decoupling loss, to resolve semantic ambiguity issues arising from inconsistent naming and hierarchical dependencies in open-vocabulary segmentation of RS images. An et al. [38] presented a soft guidance framework that includes a generalization compensation strategy, a soft guidance optimization mechanism, and multiscale contrastive learning innovations to overcome the challenges of model collapse toward known classes and difficulty in perceiving specific semantic cues in open-vocabulary segmentation of RS images. Wang et al. [39] proposed the T2ASeg method, which integrates a text-guided adaptive activation module and a multiscale structural semantic enhancement module, to address the problems of inaccurate weakly supervised pseudo-labels and noise interference during the learning process in open-vocabulary segmentation of RS images. Ye et al. [40] proposed the GSNet method, which includes a dual-stream image encoder and a residual information-preserving decoder, to address the challenges of lacking large-scale dedicated datasets for open-vocabulary segmentation of RS images and the insufficient feature extraction of general models within the domain. Zhang et al. [41] proposed the TPOV-Seg method, which includes an RS-specific text templater, text-aware feature enhancement, and a dual-branch reconstruction module, to address the issues of limited generalization of fixed prompt words and difficulty in pixel-level localization in OVSS for RS.

III. Methodology

As illustrated in Fig. 2, CDSeg primarily incorporates a DDFCM [Fig. 2(a)] that encodes features from both the natural domain and the RS domain to facilitate complementary knowledge between these domains, a CFFIM [Fig. 2(d)] that enhances the structure-category relationship, and a wavelet-cross-attention-enhanced decoder [Fig. 2(e)] that integrates dual-domain features with scale optimization.

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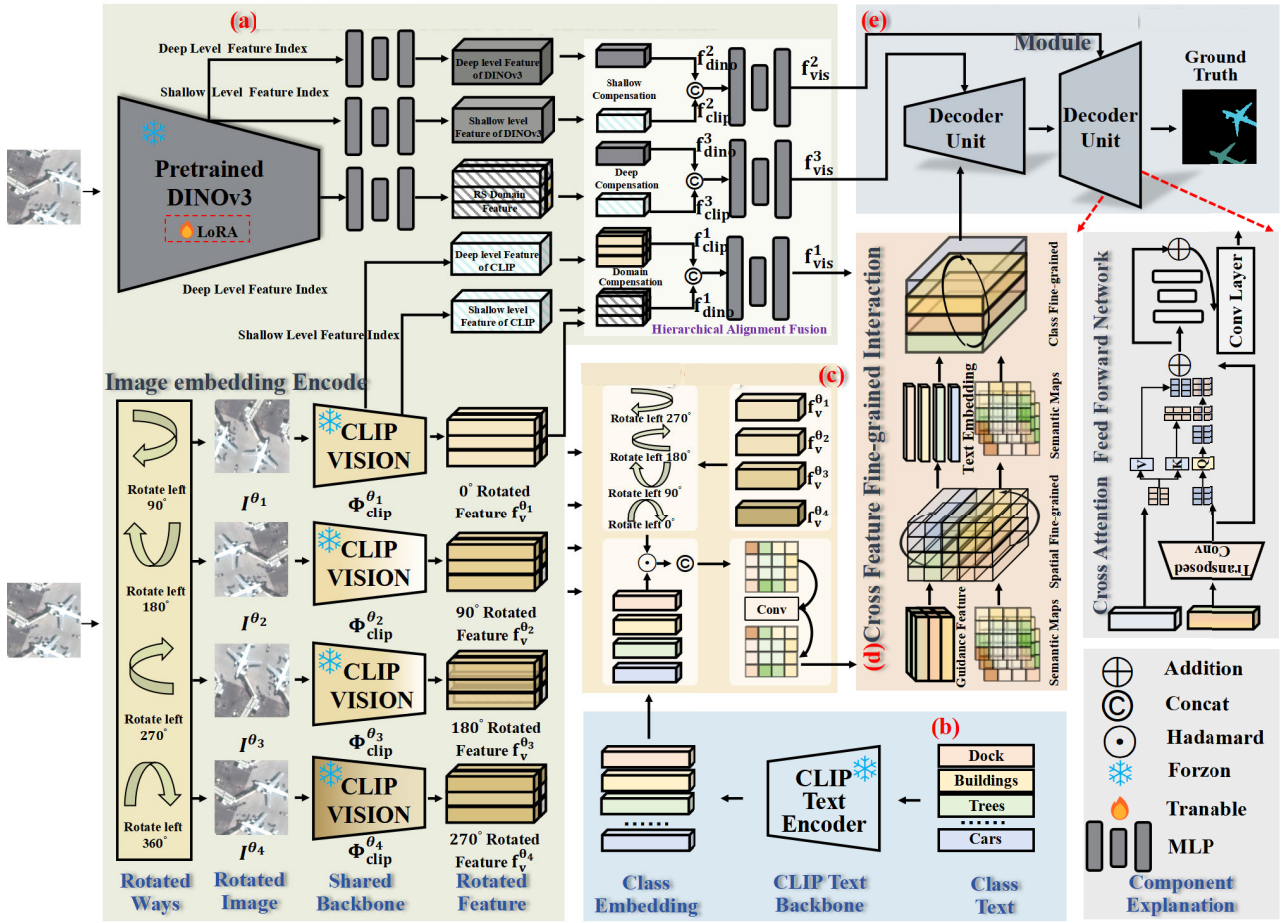


Fig. 2. Diagram shows the detailed structural design of CDSeg, including five detailed structures: DDFCM, text embedding encode, similarity computation, wavelet-cross-attention enhance module [labeled from (a) to (e)] and detailed annotations of the modules.

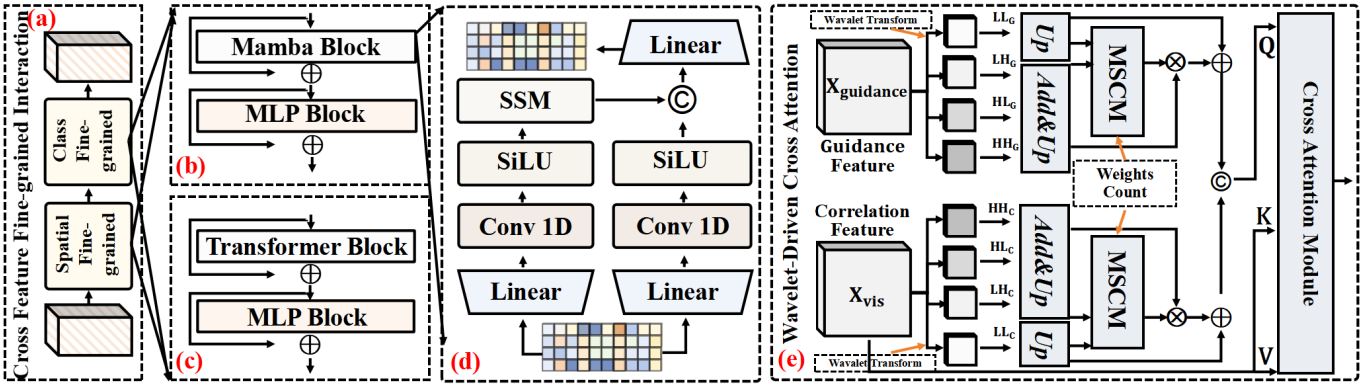


Fig. 3. Detailed structural design of CFFIM: (a) overall architecture of the cross-feature fine-grained interaction module (CFFIM), illustrating its two main branches. (b) Structure of the mamba branch, featuring cascaded Mamba and MLP blocks for sequence modeling. (c) Structure of the Transformer branch, consisting of alternating Transformer and MLP blocks. (d) Design of the wavelet-based channel attention and enhancement module (WCAEM), which utilizes frequency-domain decomposition, state space model (SSM), convolution, and channel attention mechanisms. (e) Overall workflow of the wavelet-driven cross-attention mechanism, showing multi-scale feature fusion and the final cross-attention decoding process.

A. Dual-Domain Feature Compensation Module

Traditional single-view representations frequently struggle to capture invariant features under intricate spatial layouts. We delineate a set of rotation operators $\{\mathbf{Rot}(.,\theta,i)|\theta_i \in ((\pi/2),\pi,\ldots)\}$, which executes multiangle spatial resampling on the original input image $I \in R^{3\times H\times W}$ to generate a cluster of images with diverse orientations. These images are subsequently fed into the corresponding frozen visual foundation model CLIP image encoder vit-b $\Phi_{clip}^{\theta,i}$, aiming to disentangle multiview orientation-invariant visual features $\mathbf{f}_v^{\theta,i} \in R^{512\times 24\times 24}$ in a high-dimensional latent space. To further address the challenge of varying object scales in RS images, we implement a hierarchical multiscale aggregation strategy during feature extraction. This explicitly preserves shallow texture features, mid-level structural features, and deep

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semantic features along the unrotated direction to get multilevel natural domain features $\mathbf{f}^{j}_{clip} \in R^{c_j \times ((24/2^{-j+1}) \times (24/2^{-j+1}))}$, where $j \in \{1, 2, 3\}$ denotes the feature extraction layer index and c_j represents the number of channels for features at different layers. As illustrated in Fig. 2(b), to ensure subsequent semantic alignment, we use the pretrained CLIP text encoder $\boldsymbol{\Phi}_{txt}$ to map the set of category labels $\mathbf{T}$ into the shared visual–text latent space, facilitating the extraction of text embeddings for multiple categories. The process is described by the following formula:

$$\mathbf{f}^{\theta,i}_{v}, \mathbf{f}^{j}_{clip} = \boldsymbol{\Phi}^{\theta,i}_{clip}\left(\mathbf{Rot}\left(I, \theta, i\right)\right) \tag{1}$$

$$\mathbf{f}_{txt} = \boldsymbol{\Phi}_{txt}(\mathbf{t}_j), \quad \mathbf{t}_j \in \mathbf{T} \tag{2}$$

$\mathbf{t}_j$ denotes the concise description of jth class. DDFCM aims to bridge the distribution bias between the natural domain and the RS domain through cross-domain knowledge transfer. Specifically, we map the original images into the backbone network $\boldsymbol{\Phi}_{dino}$ of DINOv3-ViT-L, which is pretrained on large-scale RS datasets such as Sat493m, to obtain multilevel RS domain features $\mathbf{f}^{j}_{dino} \in R^{c_j \times ((24/2^{-j+1}) \times (24/2^{-j+1}))}$, where $j \in \{1, 2, 3\}$ represents the hierarchical index of RS domain feature extraction. This process is described by the following formula:

$$\mathbf{f}^{j}_{dino} = \boldsymbol{\Phi}_{lora}(\boldsymbol{\Phi}_{dino}(I)). \tag{3}$$

To ensure that visual features at different hierarchical scales can comprehensively integrate dual-domain knowledge and facilitate the subsequent decoding process, we developed a *hierarchical alignment fusion* strategy. For the multilevel natural domain features and RS domain features at corresponding hierarchical levels, they are mapped into a unified canonical feature space through learnable projection operators $\boldsymbol{\omega}_1$ and $\boldsymbol{\omega}_2$, along with a concatenation operation $[\cdot]$, to eliminate dimensional inconsistencies and unify the feature distributions. This *hierarchical alignment fusion* process is described by the following formula:

$$\mathbf{f}^{j}_{vis} = \boldsymbol{\omega}_2 \cdot \left[\mathbf{f}^{j}_{clip}, \boldsymbol{\omega}_1 \cdot \mathbf{f}^{j}_{dino}\right] \tag{4}$$

$\mathbf{f}^{j}_{vis}$ represents final hierarchical fused feature. As illustrated in Fig. 2(c), to augment the deep semantic alignment between RS image content and category text, we propose a *similarity computation module*. First, a set of rotation operators $\{\mathbf{Rot}(., 2\pi - \theta, i)\}$ is utilized to calibrate multiview visual features $\mathbf{f}^{\theta,i}_{clip}$ to a global canonical direction. Then, through feature concatenation $[\cdot]$ and a cosine similarity function $\boldsymbol{\Phi}_{sim}$, the semantic correlation features $\mathbf{f}_{corr}$ between the pixel-level embeddings of multiview image features and the text embeddings $\mathbf{f}_{txt}$ are derived. The formal expression of this process is as follows:

$$\mathbf{f}_{corr} = \boldsymbol{\Phi}_{sim}\left(\left[\mathbf{Rot}\left(\mathbf{f}^{\theta,i}_{v}, 2\pi - \theta, i\right), \ldots\right], \mathbf{f}_{txt}\right). \tag{5}$$

The similarity calculation formula is defined as

$$\boldsymbol{\Phi}_{sim}\left(\mathbf{h}_v\left[q\right], \mathbf{h}_t\left[c\right]\right) = \frac{\left\langle \mathbf{h}_v\left[q\right] \cdot \mathbf{h}_t\left[c\right]\right\rangle}{\left\|\mathbf{h}_v\left[q\right]\right\| \left\|\mathbf{h}_t\left[c\right]\right\|}. \tag{6}$$

In the above formula, $\mathbf{h}_v[q]$ denotes the 2-D spatial feature at index q in the image embedding and $\mathbf{h}_t[c]$ signifies the text feature vector at index c in the text embedding. Subsequently, to refine the semantic correlation features $\mathbf{f}_{corr}$ from the dual dimensions of spatial structure and semantic categories, we propose the CFFIM $\boldsymbol{\Phi}_{cffim}$. Guided by visual semantic representations $\mathbf{f}^{l}_{vis}$ and textual embeddings $\mathbf{f}_{txt}$, this module enhances the discriminative power of the correlation features through a two-stage global modeling mechanism. This refinement process yields the refined correlation features $\hat{\mathbf{f}}_{corr}$, thereby significantly bolstering the model's generalization robustness across unseen scenarios and novel categories. The formal expression of this process is as follows:

$$\hat{\mathbf{f}}_{corr} = \boldsymbol{\Phi}_{cffim}\left(\mathbf{f}_{corr}, \mathbf{f}^{l}_{vis}, \mathbf{f}_{txt}\right). \tag{7}$$

Finally, to effectively integrate dual-domain knowledge and fully exploit the complementary properties of multidirectional and multiscale features, the refined correlation features $\hat{\mathbf{f}}_{corr}$ and hierarchical fused features $\mathbf{f}^{j}_{vis}$ are fed into the wavelet-cross-attention-enhanced decoder $\boldsymbol{\Phi}_{wcaed}$. This module dynamically computes the attention weights of $\hat{\mathbf{f}}_{corr}$ relative to visual information across different scales by leveraging wavelet transform and cross-attention mechanisms, which ultimately generates the final prediction p_{gt}. The formulation is as follows:

$$\mathbf{p}_{gt} = \boldsymbol{\Phi}_{wcaed}\left(\hat{\mathbf{f}}_{corr}, \mathbf{f}^{j}_{vis}\right). \tag{8}$$

B. Cross-Feature Fine-Grained Interaction Module

As illustrated in Fig. 3(a), unlike conventional methods for augmenting the generalization ability of fine-grained features, CFFIM innovatively proposes a unified hybrid modeling architecture that decouples feature interaction into two sequential stages: Spatial fine-grained (SF) $\boldsymbol{\Phi}_{sf}$ and class fine-grained (CF) $\boldsymbol{\Phi}_{cf}$. In the first stage (SF), we concatenate the semantic correlation features $\mathbf{f}_{corr}$ with deep guidance hierarchical fused features $\mathbf{f}^{l}_{vis}$, then utilize the alternating modeling mechanism of Mamba layers and Transformer layers within the MambaVision [42] block to enhance pixel-level perception of the visual spatial domain, thereby generating SF features $\mathbf{f}^{s}_{corr}$. In the second stage (CF), we introduce class text guidance features $\mathbf{f}_{txt}$ and apply a *repeat* operation to expand their dimensions, aligning them spatially with $\mathbf{f}^{s}_{corr}$. After concatenation and fusion, the combined features are fed back into the MambaVision module for secondary modeling, ultimately producing highly discriminative fine-grained semantic correlation features $\hat{\mathbf{f}}_{corr}$. This process effectively promotes the model's generalization capacity in unseen scenarios through cross-modal feature interaction. The formal expression of this process is as follows:

$$\hat{\mathbf{f}}_{corr} = \boldsymbol{\Phi}_{cf}\left(\left[\boldsymbol{\Phi}_{sf}\left(\left[\mathbf{f}_{corr}, \mathbf{f}^{l}_{vis}\right]\right), \mathbf{Repeat}\left(\mathbf{f}_{txt}\right)\right]\right). \tag{9}$$

To effectively capture global context and establish long-range dependencies between vision and text features, CFFIM adopts a specialized Mamba–Transformer hybrid architecture across two stages, which is the core of CFFIM. As illustrated in Fig. 3(b) and (c), the core of global modeling follows a sequential design of "*Mamba block first, then Transformer block.*" Specifically, for the guidance feature $\mathbf{f}_k$ at any stage (where $k \in \{vis, txt\}$), the module first concatenates it with

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the semantic correlation feature $\mathbf{f}_{\text{corr}}$ and normalizes $\boldsymbol{\Phi}_{\text{norm}}$ the result. Then, the feature flows through the global modeling core $\boldsymbol{\Phi}_{\text{attn}}$ (i.e., Mamba or Transformer) and the MLP module $\boldsymbol{\Phi}_{\text{mlp}}$. The internal computation process is described by the following formula:

$$\mathbf{f}_{\text{tmp}} = \boldsymbol{\Phi}_{\text{attn}}\left(\boldsymbol{\Phi}_{\text{norm}}\left([\mathbf{f}_{\text{corr}}, \mathbf{f}_{\text{k}}]\right)\right) + [\mathbf{f}_{\text{corr}}, \mathbf{f}_{\text{k}}] \tag{10}$$

$$\mathbf{f}_{\text{out}} = \boldsymbol{\Phi}_{\text{mlp}}\left(\boldsymbol{\Phi}_{\text{norm}}\left(\mathbf{f}_{\text{tmp}}\right)\right) + \mathbf{f}_{\text{tmp}} \tag{11}$$

$\mathbf{f}_{\text{tmp}}$ and $\mathbf{f}_{\text{out}}$ represent the intermediate process features and output features of the current module, respectively. As shown in Fig. 3(c), one of the core components of global modeling is the Mamba architecture. The main branch primarily uses SSM $\boldsymbol{\Phi}_{\text{ssm}}$ to perform global context modeling of guided features and associated features, facilitating the understanding of associated features in terms of spatial vision and category semantics. The auxiliary branch mainly employs 1-D convolution operation $\boldsymbol{\kappa}_{\mathbf{1\times 1}}$ and the sigmoid linear unit (SiLU) activation function δ to compensate for local details or noncausal information that may be lost due to the sequential modeling nature of SSM. This process is described by the following formula:

$$\mathbf{f}_{\text{tmp}} = \delta\left(\boldsymbol{\kappa}_{\mathbf{1\times 1}}\left(\omega_3 \cdot [\mathbf{f}_{\text{corr}}, \mathbf{f}_{\text{k}}]\right)\right) \tag{12}$$

$$\mathbf{f}_{\text{out}} = \left[\boldsymbol{\Phi}_{\text{ssm}}\left(\mathbf{f}_{\text{tmp}}\right), \delta\left(\boldsymbol{\kappa}_{\mathbf{1\times 1}}\left(\omega_4 [\mathbf{f}_{\text{corr}}, \mathbf{f}_{\text{k}}]\right)\right)\right]. \tag{13}$$

The discretized SSM [43] is the core of the Mamba framework. Inspired by linear time-invariant (LTI) systems, for a 1-D input sequence $\mathbf{x}_t$, the SSM processes the data using the discretized form of the following ordinary differential equations (ODEs) by introducing discretization parameters $\overline{\mathbf{A}}$, $\overline{\mathbf{B}}$, $\overline{\mathbf{C}}$, and an intermediate hidden state $\mathbf{h}_{t-1}$, to obtain the final output $\mathbf{y}_t$:

$$\mathbf{h}_{\text{t}} = \overline{\mathbf{A}}\mathbf{h}_{\text{t}-1} + \overline{\mathbf{B}}\mathbf{x}_{\text{t}} \tag{14}$$

$$\mathbf{y}_{\text{t}} = \overline{\mathbf{C}}\mathbf{h}_{\text{t}}. \tag{15}$$

The core of the Transformer block in global modeling remains the original multihead attention mechanism $\boldsymbol{\Phi}_{\text{multihead}}$, which is described by the following formula:

$$\mathbf{Q},\ \mathbf{K},\ \mathbf{V} = (\omega_5, \omega_6, \omega_7) \cdot \mathbf{x}_{\text{in}} \tag{16}$$

$$\boldsymbol{\Phi}_{\text{multihead}}\left(\mathbf{Q},\ \mathbf{K},\ \mathbf{V}\right) = \left(\frac{\mathbf{Q}\cdot\mathbf{K}}{\sqrt{\mathbf{d}}}\right)\cdot\mathbf{V} \tag{17}$$

$\mathbf{x}_{in}$ represents the input features of the current module and d represents the feature dimension.

C. Wavelet-Cross-Attention-Enhanced Module

To bridge the semantic gap and preserve high-frequency spatial details, we propose the WCAEM, which is structurally composed of stacked decoder units that internally integrate the wavelet cross-attention mechanism. As illustrated in Fig. 3(e), WCAEM constitutes the core of our decoder, specifically engineered to facilitate scale-aware attention of fine-grained correlation features $\hat{\mathbf{f}}_{\text{corr}}$ toward the multiscale hierarchical fused features $\mathbf{f}^{\text{j}}_{\text{vis}}$. Since high-frequency subbands more prominently highlight the target's subtle scale characteristics, we first employ wavelet transform $\boldsymbol{\Phi}_{\text{wt}}$ from the frequency-domain perspective to decompose $\hat{\mathbf{f}}_{\text{corr}}$ and $\mathbf{f}^{\text{j}}_{\text{vis}}$ into four components with half the pixel resolution: $\mathbf{HH}_{\text{k}}$, $\mathbf{HL}_{\text{k}}$, $\mathbf{LH}_{\text{k}}$, $\mathbf{LL}_{\text{k}}$, and $\text{k} \in \{\text{corr}, \text{vis}\}$. $\mathbf{HH}_{\text{k}}$, $\mathbf{HL}_{\text{k}}$, and $\mathbf{LH}_{\text{k}}$ denote high-frequency details in the horizontal, vertical, and diagonal directions, respectively. To comprehensively utilize scale characteristics, we sum the high-frequency parts element-wise to obtain the integrated high-frequency feature $\mathbf{f}^{\text{k}}_{\text{h}}$, while the low-frequency part $\mathbf{f}^{\text{k}}_{\text{l}}$ remains unchanged. Finally, both are fed back into the multiscale context module (MSCM) $\boldsymbol{\Phi}_{\text{mscm}}$ to generate multiscale channel weights [44]. These weights are applied to the high-frequency components, then fused with the low-frequency residual components through a skip connection to obtain the frequency-domain multiscale representation $\mathbf{s}^{\text{k}}_{\text{h}}$ of the fine-grained correlation feature $\hat{\mathbf{f}}_{\text{corr}}$ and the guidance feature $\mathbf{f}^{\text{j}}_{\text{vis}}$. This process is described by the following formula:

$$\mathbf{HH}_{\text{k}}, \mathbf{HL}_{\text{k}}, \mathbf{LH}_{\text{k}}, \mathbf{LL}_{\text{k}} = \boldsymbol{\Phi}_{\text{wt}}\left(\mathbf{f}_{\text{k}}\right) \tag{18}$$

$$\mathbf{f}^{\text{k}}_{\text{h}}, \mathbf{f}^{\text{k}}_{\text{l}} = \mathbf{HH}_{\text{k}} + \mathbf{HL}_{\text{k}} + \mathbf{LH}_{\text{k}}, \mathbf{LL}_{\text{k}} \tag{19}$$

$$\mathbf{s}^{\text{k}}_{\text{h}} = \boldsymbol{\Phi}_{\text{mscm}}\left(\mathbf{f}^{\text{k}}_{\text{h}}, \mathbf{f}^{\text{k}}_{\text{l}}\right)\cdot\mathbf{f}^{\text{k}}_{\text{h}} + \mathbf{f}^{\text{k}}_{\text{l}}. \tag{20}$$

To enable $\hat{\mathbf{f}}_{\text{corr}}$ to emphasize frequency-domain detailed features, we employ a cross-attention mechanism to concatenate the multiscale frequency-domain features $\mathbf{s}^{\text{corr}}_{\text{h}}$ and $\mathbf{s}^{\text{vis}}_{\text{h}}$ as the query, while utilizing $\hat{\mathbf{f}}_{\text{corr}}$ as the key and value to determine the attention level. This process is described by the following formula:

$$\mathbf{f}_{\text{tmp}} = \boldsymbol{\Phi}_{\text{multihead}}\left(\left[\mathbf{s}^{\text{vis}}_{\text{h}}, \mathbf{s}^{\text{corr}}_{\text{h}}\right], \hat{\mathbf{f}}_{\text{corr}}, \hat{\mathbf{f}}_{\text{corr}}\right) + \hat{\mathbf{f}}_{\text{corr}} \tag{21}$$

$$\mathbf{f}_{\text{out}} = \boldsymbol{\Phi}_{\text{mlp}}\left(\boldsymbol{\Phi}_{\text{norm}}\left(\mathbf{f}_{\text{tmp}}\right)\right) + \mathbf{f}_{\text{tmp}} \tag{22}$$

$\mathbf{f}_{\text{tmp}}$ and $\mathbf{f}_{\text{out}}$ represent the intermediate process features and output features of the current module, respectively.

IV. Experiments and Analysis

A. Dataset Description

1) ISPRS Potsdam: Potsdam is a highly representative high-resolution benchmark dataset in the field of urban scene semantic segmentation. This dataset consists of 24 ultrahigh-resolution true orthophotos with a spatial resolution of 0.05 m. Each image has a spatial size of up to 6000×6000 pixels. In terms of data modalities, Potsdam provides four-channel data including near-infrared, red, green, and blue (IRRGB), offering rich multispectral clues for detailed analysis of complex urban scenes. Its pixel-level ground truth annotations cover six typical land cover classes: buildings, trees, low vegetation, vehicles, impervious surfaces, and clutter.

2) ISPRS Vaihingen: This dataset also focuses on semantic parsing of complex urban scenes and contains a total of 16 high-resolution orthophotos. Each image has spatial dimensions of 2500×2000 pixels. In terms of data modalities, the dataset provides three-channel false-color spectral information in near-infrared, red, and green (IRRG), offering highly distinctive multispectral features for precise segmentation of urban objects. Its semantic category space is consistent with that of the Potsdam dataset.

3) DLRSD Dataset [45]: The Dense Labeling RS Dataset is a classic benchmark dataset for dense prediction tasks in RS imagery. This dataset is an extension built upon the UC Merced Land Use dataset and contains a total of 7002 high-resolution RS images with a spatial resolution of

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TABLE I
PERFORMANCE COMPARISONS OF DIFFERENT METHODS WITH DIFFERENT DATASETS AS THE TRAINING DATASETS. VAL DATA DENOTES THE VALIDATION DATASET OF THE CHOSEN TRAINING DATASET

Method	DLRSD			iSAID			Potsdam			Vaihingen		
	mIoU	fwIoU	mACC	mIoU	fwIoU	mACC	mIoU	fwIoU	mACC	mIoU	fwIoU	mACC
DLRSD as training dataset.												
SCAN	48.52	55.01	68.68	34.18	44.62	49.05	20.22	28.37	34.70	5.38	7.72	22.54
SAN	85.73	86.24	91.03	30.63	39.87	44.03	**30.30**	**38.04**	**44.98**	31.92	**40.23**	45.36
FreeSeg	85.46	86.39	91.39	30.13	40.14	41.49	20.87	28.16	18.57	28.82	29.77	43.37
GKC	85.24	86.09	90.54	29.95	39.86	39.98	22.47	27.16	38.67	30.18	36.61	46.37
SED	85.13	86.80	91.36	21.54	25.72	36.28	19.47	21.29	33.40	29.40	36.77	49.38
CAT-Seg	85.84	86.79	91.44	23.56	27.40	38.48	26.79	31.28	44.72	32.32	38.71	49.65
OVRS	85.98	86.94	91.52	39.09	**48.97**	54.43	27.47	26.81	42.07	33.71	38.57	**50.01**
Ours	**87.74**	**88.19**	**92.94**	**39.26**	48.35	**55.01**	28.74	26.56	44.88	**35.14**	39.30	49.76
iSAID as training dataset.												
SCAN	17.09	19.50	41.11	60.37	62.41	74.43	19.41	25.91	40.17	9.88	13.16	26.40
SAN	20.28	19.20	**47.95**	85.67	88.02	90.99	14.79	17.69	36.51	16.38	17.53	35.38
FreeSeg	20.68	20.79	43.81	91.81	91.68	94.31	17.67	21.23	32.87	13.45	12.81	28.45
GKC	24.75	23.89	43.97	91.89	91.93	94.87	16.48	20.16	31.12	13.03	10.31	32.68
SED	25.45	24.09	44.54	92.92	92.16	95.49	15.27	18.96	28.57	12.82	9.61	29.37
CAT-Seg	22.13	23.93	44.21	92.33	92.51	95.19	14.68	14.40	37.47	9.93	10.87	38.40
OVRS	20.67	**26.18**	45.44	93.29	92.52	95.77	19.94	26.29	49.12	20.82	26.61	50.60
Ours	**26.03**	25.64	44.98	**94.25**	**93.76**	**96.00**	**21.04**	**27.39**	**51.54**	**22.46**	**27.83**	**51.34**

256×256 pixels. It covers 17 fine-grained semantic classes. Given the significant variation in target scales and the highly complex backgrounds in RS scenes, this dataset can effectively test a model's fine-grained perception capabilities in complex domains.

4) ISAID Dataset [46]: This dataset is a highly challenging large-scale benchmark in the field of RS visual analysis. It contains a total of 24 439 high-resolution aerial images, covering 15 typical object categories. Unlike conventional natural scene images, the objects in iSAID exhibit extreme scale variations, arbitrary directional distributions, and highly dense arrangements. This provides an excellent benchmark for evaluating a model's fine-grained feature alignment and perception capabilities in complex open environments.

B. Implementation Details

During the training stage, the model is optimized utilizing an AdamW optimizer with an initial learning rate of 0.0002 alongside a weight decay of 0.0001 to mitigate overfitting, while the learning rate scheduler is WarmupCosineLR. The max-iter number and batch size are set to 80 000 and 2, respectively. All models are trained on a single GTX 4090 GPU (24G) using PyTorch 2.7.0 and Python 3.12.3. Furthermore, to better adapt to CLIP's visual feature encoder, we resized all images to a uniform size of $3 \times 384 \times 384$. During the experiments, CDSeg was trained independently on the iSAID and DLRSD datasets and subsequently tested on the remaining three datasets. Finally, our workstation is equipped with an Intel[1] Core[2] i9-14900KF CPU and runs the Ubuntu 20.04 operating system. For more detailed hyperparameter settings, refer to the YAML files in our code. Specifically, we would like to express our sincere gratitude to the authors of OVRS for their foundational contributions, which served as a significant inspiration for the conceptualization and development of our framework.

[1]Registered trademark.
[2]Trademarked.

C. Loss Function

For the loss function, we adopt the Dice loss as the sole supervision signal throughout training. Given the predicted probability p_i at pixel i and the corresponding ground-truth label y_i, with N pixels in total, the Dice coefficient is as

$$\text{Dice} = \frac{2\sum_{i=1}^{N} y_i p_i}{\sum_{i=1}^{N} y_i + \sum_{i=1}^{N} p_i}. \tag{23}$$

The Dice loss is calculated as 1 − Dice and is applied to supervise both the pseudo labels and the final predictions from the decoder. This approach encourages maximal overlap between predicted and reference regions, and is robust to the foreground-background imbalance.

D. Evaluation Metrics

To comprehensively assess the segmentation performance of the proposed model, three widely adopted evaluation metrics were employed: frequency-weighted intersection over union (fwIoU), mean intersection over union (mIoU), and mean accuracy (mAcc). These metrics are established standards in the field of RS image segmentation. The fwIoU metric takes into account the frequency of each category's occurrence in the data to weight the calculation of the intersection over union

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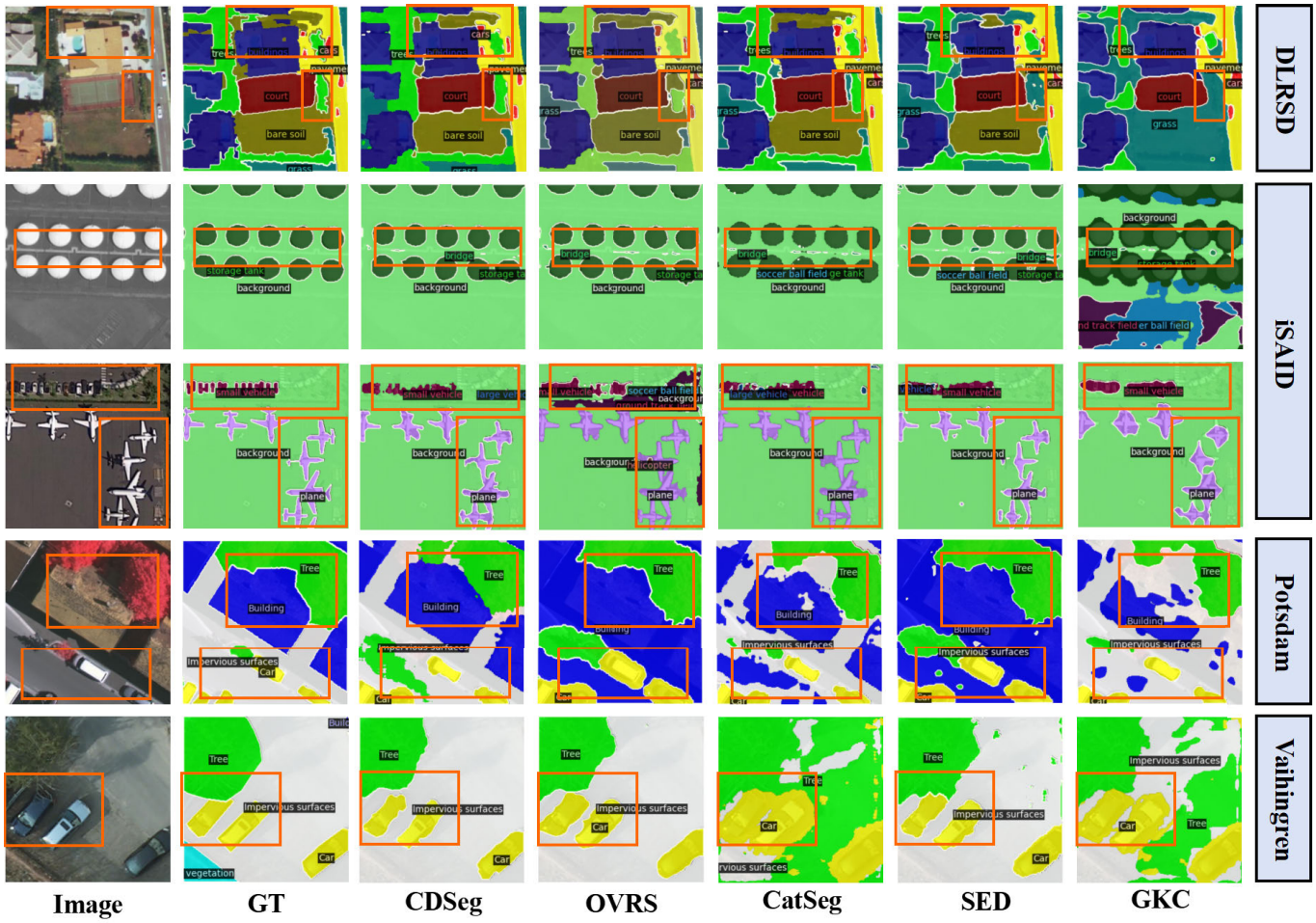


Fig. 4. Qualitative comparison of segmentation results on the four datasets trained on the DLRSD dataset. The visualization presents representative urban scenarios, including plane, car, and tree, where the results of the proposed CDSeg are compared against four SOTA methods: OVRS, CatSeg, SED, and GKC. The examples highlight the superior localization accuracy, semantic consistency, and robustness of CDSeg in handling complex scenes.

(IoU). The advantage is that it does not focus only on frequently occurring categories while ignoring those that appear less often. The metric is defined by the following formula:

$$\text{fwIoU} = \frac{1}{\sum_{i=0}^{k} t_i} \sum_{i=0}^{k} \frac{t_i \cdot p_{ii}}{t_i + \sum_{j=0}^{k} p_{ji} - p_{ii}}. \tag{24}$$

Here, p_{ii} represents the total number of pixels whose true class is class i and are correctly predicted as class i by the model; p_{ji} represents the total number of pixels whose true class is class j but are incorrectly predicted as class i by the model; t_i represents the total number of pixels in the dataset that truly belong to class i; and k denotes the total number of classes. In the open vocabulary task, mAcc evaluates the model's classification accuracy for each class by calculating the proportion of correctly classified pixels within each class, and then averaging the accuracy across all categories. Its calculation formula is

$$\text{mAcc} = \frac{1}{k+1} \sum_{i=0}^{k} \frac{p_{ii}}{t_i}. \tag{25}$$

In contrast, the mIoU emphasizes fairness across categories, reflecting the model's average segmentation capability among different classes. It quantifies the degree of overlap between the predicted and annotated regions at the class level and is formulated as follows:

$$\text{mIoU} = \frac{1}{|C|} \sum_{c \in C} \frac{|R_c \cap T_c|}{|R_c \cup T_c|}. \tag{26}$$

In this formula, $|C|$ denotes the set of categories, and R_c and T_c represent the predicted and ground-truth pixel sets for category c, respectively.

E. Comparison With SOTA Methods

To systematically evaluate the effectiveness and generalization capability of CDSeg, this study selected representative models from the field of RSOVSS for a comprehensive comparison. The chosen benchmark methods include OVRS [37], Catseg [47], SED [28], GKC [48], FreeSeg [49], SAN [50], and SCAN [51]. This overall comparison framework balances technological stages and methodological innovation, fully reflecting the technological evolution over the past decade. Under a strict cross-domain evaluation setting, we conducted extensive quantitative comparisons between our model and the current state-of-the-art (SOTA) OVSS methods. Table I presents the overall evaluation results of our method after training on the DLRSD and iSAID datasets, followed by zero-shot transfer to multiple heterogeneous datasets.

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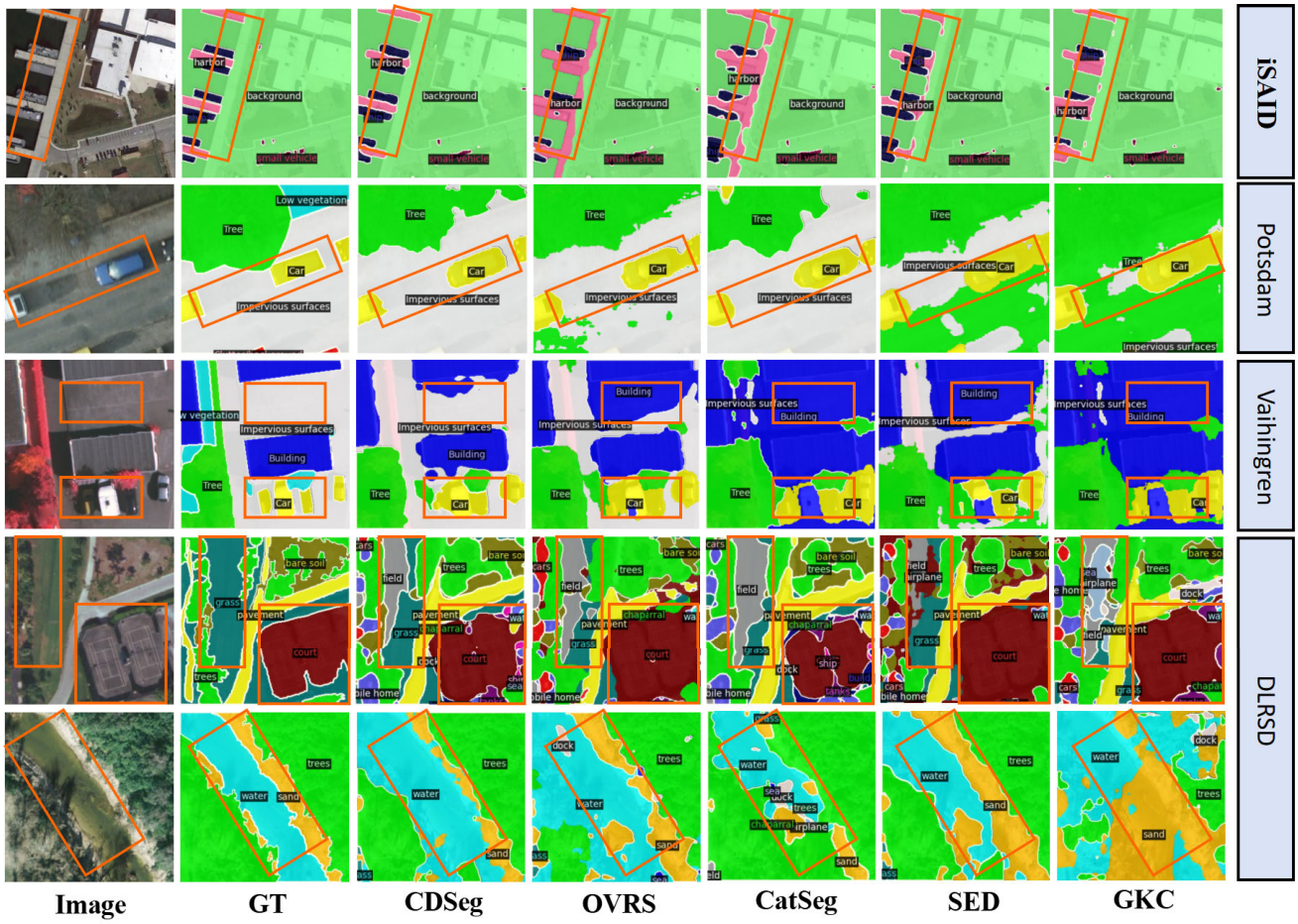


Fig. 5. Qualitative comparison of segmentation results on the four datasets trained on the iSAID dataset. The visualization covers a variety of challenging RS scenes, such as small bridges, planes, and vehicles. The proposed VLF-BG is compared with four leading methods (FIANet, ReferSAM, Enti-TwistNet, and DANet), demonstrating its ability to produce precise boundaries and maintain semantic consistency.

1) Performance Comparison on the Training DLRSD Dataset: After training on the DLRSD dataset, our method demonstrates outstanding cross-domain generalization ability across the vast majority of metrics. Specifically, it achieves an mIoU of 87.74% on the source validation set, which is a 1.76% improvement over the previous best model, OVRS, fully showcasing its capability in representing complex fine-grained targets and alignment accuracy. On the target domains iSAID and Vaihingen, the mIoUs are 39.26% and 35.14%, respectively, both surpassing the previous best model OVRS. It is worth noting that although our method's mIoU on the Potsdam dataset is 28.74%, slightly lower than SAN's 30.30%, it still outperforms the strongest baseline OVRS at 27.47%. Meanwhile, SAN's performance significantly degrades on the more challenging iSAID dataset, which features greater scale variation and more complex categories, achieving only 30.63% mIoU. In contrast, our method benefits from the DDFCM module's robust extraction of structural features, effectively avoiding overfitting to any single specific domain distribution, and maintains the highest overall performance across all test scenarios.

2) Scenarios Comparison on the Training DLRSD Dataset: As illustrated in Fig. 4, to intuitively verify the cross-domain generalization ability and fine-grained segmentation performance of the proposed method, we present qualitative visual comparison results on heterogeneous target domains iSAID, Potsdam, and Vaihingen, using DLRSD as the source-domain training set. From the overall visual perspective, compared to other SOTA models, the segmentation masks predicted by our CDSeg are closest to the ground truth in terms of semantic consistency and boundary adherence, effectively overcoming common issues of semantic confusion and object fragmentation in cross-domain scenarios. Specifically, when dealing with fine-grained small objects and densely distributed scenes, such as the small airplanes in the third row of iSAID and the cars in the fifth row of Vaihingen in Fig. 4, methods like SED and GKC suffer from severe missed detections or overly smoothed boundaries. In contrast, CDSeg accurately captures the geometric contours of objects by extracting and enhancing high-frequency edge information and local detail features, enabling the model to maintain sensitive perception of tiny targets even when the resolution or scale changes in the target domain. When transferring from the source domain to target domains with drastically different land cover styles, lighting conditions, and other characteristics, such as the buildings in the fourth row of Potsdam and the bridges in the second row of iSAID, OVRS, and CatSeg tend to overfit to specific appearances of the source domain.

3) Performance Comparison on the Training iSAID Dataset: When using the more challenging iSAID dataset as the training set, our method once again demonstrates

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TABLE II
PERFORMANCE COMPARISONS OF DIVERSE CLASSES ON THE VAIHINGEN DATASET. THE BEST RESULTS ARE SHOWN IN BOLD

Methods	Impe. Sur.	Building	Low vegetation	Tree	Car	background	mean IoU
DLRSD as the training dataset							
SCAN	13.69	13.93	0.86	0.23	2.51	1.06	5.38
SAN	**42.59**	58.50	14.75	41.50	34.03	0.15	31.92
FreeSeg	37.19	54.75	13.41	37.18	28.23	2.13	28.82
GKC	39.09	55.13	14.75	38.06	32.19	1.89	30.18
SED	41.29	50.75	**15.36**	36.40	29.43	3.15	29.40
CAT-Seg	37.54	65.22	9.20	38.05	36.71	7.19	32.32
OVRS	37.27	66.07	2.50	42.35	44.77	**9.29**	33.71
Ours	40.42	**68.95**	4.77	**44.80**	**46.08**	5.83	**35.14**
iSAID as the training dataset							
SCAN	14.54	15.92	6.81	14.93	5.87	1.24	9.88
SAN	21.65	0.62	**21.34**	28.22	25.17	1.29	16.88
FreeSeg	27.13	6.75	2.13	21.23	21.75	1.75	13.45
GKC	25.46	10.75	4.76	19.40	16.75	1.09	13.03
SED	**28.79**	0.30	4.48	0.12	**41.26**	1.95	12.82
CAT-Seg	26.28	1.40	2.83	10.17	15.99	0.93	9.93
OVRS	21.79	**36.69**	2.80	43.85	16.23	3.56	20.82
Ours	26.57	35.43	3.03	**45.49**	19.36	**4.88**	**22.46**

TABLE III
EFFICIENCY COMPARISONS BETWEEN DIVERSE SOTA METHODS

Methods	Params. (M)	GFLOPs
SCAN	593.2	857.2
SAN	8.4	64.3
FreeSeg	349.2	1537.6
GKC	40.5	151.4
SED	90.7	254.3
CAT-Seg	433.7	2121.4
OVRS	447.5	2626.5
CDSeg	**509.7**	**2945.3**

excellent cross-domain generalization ability. Our model achieves mIoU, fwIoU, and mACC of 94.25%, 93.76%, and 96.00%, respectively, on the iSAID source validation set, comprehensively outperforming the previous best model OVRS. More importantly, when dealing with target domains with large disparities such as Potsdam and Vaihingen, thanks to the CFFIM module's ability to extract robust features with high domain invariance and align complementary information across different modalities, our method still demonstrates excellent performance on unseen target domains, achieving mIoU scores of 21.04% and 22.46%, respectively, both surpassing the previous SOTA model OVRS.

4) Scenarios Comparison on the Training iSAID Dataset: Fig. 5 shows a comparison of the model's performance in cross-domain open-vocabulary segmentation, using iSAID as the source domain and Potsdam, Vaihingen, and DLRSD as the target domains. This further demonstrates that CDSeg's cross-domain prediction quality significantly outperforms existing methods. Specifically, in the third row visualization of the Vaihingen dataset, when target domain classes have similar grayscale textures, models like OVRS and CatSeg mistakenly identify flat impermeable surfaces as buildings due to their overreliance on local visual priors from the source domain. In the fourth and fifth rows of the DLRSD dataset, when confronted with target-domain-specific land cover styles, CatSeg misclassifies tennis courts as ships or tanks, and water as airplanes or docks. In contrast, our CDSeg shows remarkable accuracy and robustness in these highly challenging target domains. When facing easily confused classes and unseen land cover appearances, CDSeg leverages the efficient collaboration of the decoupled dual-frequency context module and the cross-modal feature fusion interaction module to filter out appearance noise caused by style and lighting differences, enabling the model to still produce segmentation results that closely match the ground truth.

5) Detailed Class Comparison on Two Datasets: To thoroughly investigate performance, Table II presents a detailed breakdown of fine-grained category segmentation results of different methods on the Vaihingen dataset. Overall, whether using DLRSD or iSAID as the source-domain training set, our method achieves mIoU scores of 35.14% and 22.46%, respectively, representing improvements of 1.43% and 1.64% over the previous best-performing OVRS. In terms of specific categories, our model demonstrates significant performance gains when handling categories with large-scale variations and strong background interference during cross-domain adaptation. Specifically, when trained on the DLRSD dataset, our method attains mIoU scores of 68.95% and 46.08% for the building and car categories, respectively, substantially outperforming the previously best model, OVRS. These notable improvements highlight CDSeg's ability to effectively extract and align high-level structural semantics and spatial boundary information of targets, underscoring its importance for semantic understanding in earth observation tasks. It is worth noting that CDSeg still has shortcomings in categories such as low vegetation and impervious surfaces, which heavily

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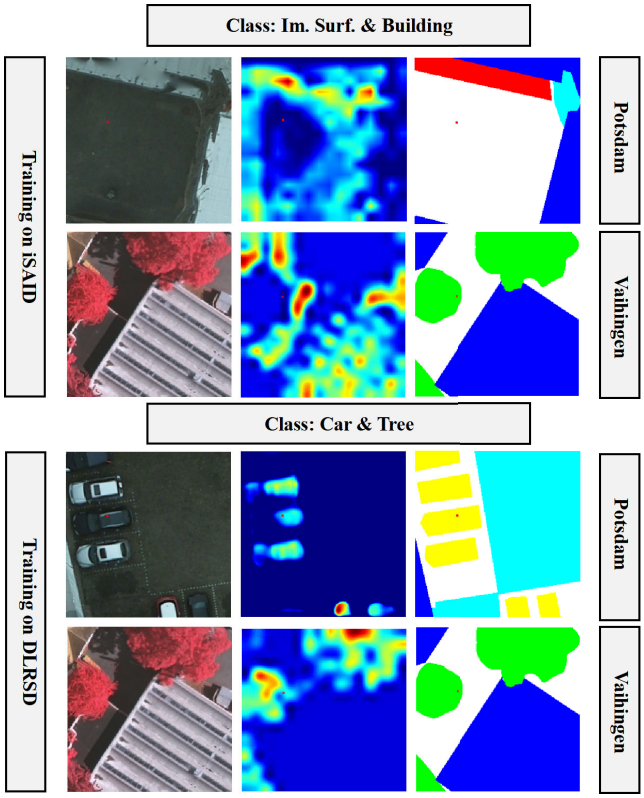


Fig. 6. Qualitative visualization of feature heatmaps from ablation experiments on the Potsdam and Vaihingen datasets. This image contains different heatmaps obtained from iSAID and DLRSD training, as well as testing on the Potsdam and Vaihingen datasets.

rely on spectral reflectance and seasonal color variations. We believe the main reason is the difference in underlying visual features captured by different sensors during the acquisition of the DLRSD and Vaihingen datasets. Although SAN achieved higher scores on low vegetation, it often overfits to the specific color distribution of the source domain, significantly weakening its ability to recognize other core categories. In contrast, our network design focuses more on learning domain-invariant structural features. From a global perspective, this strategy successfully avoids local feature collapse, thereby ensuring the model's stable and superior overall performance on unknown target domains.

To provide a more intuitive illustration of CDSeg's discriminative capabilities in cross-dataset scenarios, we present the visual heatmaps and t-SNE manifolds in Figs. 6 and 7. These visualizations were obtained by deploying the CDSeg model pretrained on iSAID and DLRSD directly onto the Potsdam and Vaihingen benchmarks for zero-shot evaluation. As observed in the heatmaps, while the activation intensity of CDSeg appears relatively diffuse compared to fully supervised closed-set baselines, it successfully captures the salient regions of target categories. This attenuation in response intensity is primarily a consequence of the domain gap inherent in cross-dataset transfer, where the model must operate on unseen data distributions without fine-tuning. Notably, CDSeg still preserves a substantial level of discriminative focus, which can be attributed to the integration of DINOv3. The robust RS-domain priors encapsulated in DINOv3 provide an explicit

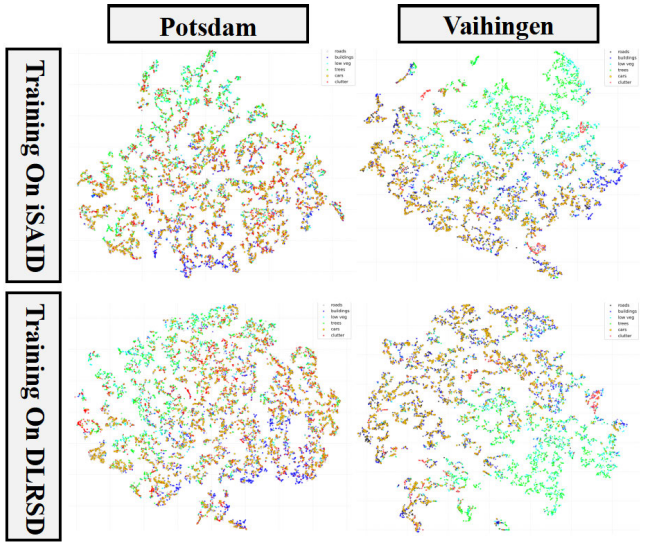


Fig. 7. Qualitative visualization of t-SNE from ablation experiments on the Potsdam and Vaihingen datasets. This image contains different t-SNE images obtained from iSAID and DLRSD training, as well as testing on the Potsdam and Vaihingen datasets.

semantic augmentation to the image features, effectively bridging the gap between heterogeneous datasets. Furthermore, the t-SNE visualizations corroborate these findings. Despite the lack of domain-specific supervision, CDSeg facilitates the formation of well-defined feature clusters with high intraclass compactness and clear interclass boundaries. This evidence further substantiates that CDSeg possesses superior feature representation and generalization capabilities, enabling it to maintain reliable semantic consistency across diverse RS environments.

F. Parameter Analysis

As illustrated in Table III, to provide a rigorous empirical assessment of the proposed CDSeg framework, we conducted a comprehensive complexity analysis focusing on two primary computational indicators: the total number of parameters (Params) and giga floating point operations (GFLOPs). CDSeg comprises 509.7 M parameters and 2945.3 GFLOPs, representing a higher computational footprint compared to contemporary open-vocabulary methods such as OVRS. This increased complexity is primarily attributable to the integration of dual high-capacity visual encoders during the feature compensation stage, as well as the deployment of wavelet-based decomposition and cross-attention mechanisms within the decoding architecture. While these components necessitate greater computational resources, the resulting architectural complexity is a justified tradeoff for the model's superior discriminative power, as evidenced by its ability to consistently achieve SOTA segmentation accuracy across multiple benchmarks.

G. Ablation Studies

As illustrated in Table IV, to rigorously evaluate the effectiveness of the core components within the proposed CDSeg model and their synergistic effects, we conducted

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TABLE IV
PERFORMANCE COMPARISONS AND ABLATION STUDY RESULTS ACROSS DIFFERENT DATASETS AND TRAINING SETTINGS

Modules			DLRSD			iSAID			Potsdam			Vaihingen		
DDFCM	**CFFIM**	**WCAEM**	mIoU	fwIoU	mACC	mIoU	fwIoU	mACC	mIoU	fwIoU	mACC	mIoU	fwIoU	mACC
iSAID for training														
✗	✗	✗	24.31	23.01	40.45	91.34	90.87	91.54	18.78	22.56	47.63	18.35	26.64	46.67
✓	✗	✗	24.56	24.87	42.40	93.37	91.45	94.70	19.82	25.61	49.92	19.95	27.00	46.82
✓	✓	✗	25.49	25.80	44.05	93.03	92.70	94.82	20.42	26.95	50.96	20.75	26.25	49.66
✗	✓	✓	25.72	24.89	44.02	92.73	92.90	95.07	20.96	**27.50**	51.00	19.82	**28.80**	50.23
✓	✗	✓	25.03	25.12	43.27	93.19	91.88	94.23	19.57	25.83	49.51	21.03	25.04	48.96
✓	✓	✗	25.49	**25.80**	44.05	93.03	92.70	94.82	20.42	26.95	50.96	20.75	26.25	49.66
✓	✓	✓	**26.03**	25.64	**44.98**	**94.25**	**93.76**	**96.00**	**21.04**	27.39	**51.54**	**22.46**	27.83	**51.34**
DLRSD for training														
✗	✗	✗	85.31	85.33	88.82	36.69	45.46	47.96	26.45	20.11	39.73	30.35	36.57	44.97
✓	✗	✗	85.82	86.98	90.93	37.99	46.09	50.59	26.27	23.34	41.39	31.02	37.97	48.83
✓	✓	✗	86.51	87.30	91.46	37.56	47.18	53.41	27.19	25.79	43.32	33.01	38.46	47.76
✗	✓	✓	86.90	87.02	90.48	38.45	46.38	54.22	27.51	25.13	43.86	33.21	38.05	48.95
✓	✗	✓	85.61	86.81	90.86	**39.78**	45.42	52.85	26.04	24.52	43.77	32.40	37.17	47.91
✓	✓	✗	86.51	87.30	91.46	37.56	47.18	53.41	27.19	25.79	43.32	33.01	38.46	47.76
✓	✓	✓	**87.74**	**88.19**	**92.94**	39.26	**48.35**	**55.01**	**28.87**	**26.56**	**44.88**	**35.14**	**39.30**	**49.76**

TABLE V
PERFORMANCE COMPARISONS ACROSS DIFFERENT DATASETS WITH TRADITIONAL METHODS AND RSOVSS METHOD

Methods	iSAID as the training dataset			DLRSD as the training dataset		
	iSAID	Potsdam	Vaihingen	DLRSD	Potsdam	Vaihingen
DeepLabv3	6.58	9.79	10.34	2.58	8.41	6.98
Segformer	7.39	10.16	13.65	2.72	8.76	7.01
ST-UNet	7.80	11.04	13.90	2.88	8.63	7.83
FTransUNet	8.32	12.41	14.78	3.02	8.78	7.69
OVRS	37.56	27.19	33.01	20.67	19.94	20.82
CDSeg	**39.26**	**28.74**	**35.14**	**26.03**	**21.04**	**22.46**

comprehensive additive and subtractive ablation studies under the cross-domain OVSS setting. Specifically, the incremental integration of our innovative modules into the Baseline consistently yields stable and significant performance gains. Integrating the DDFCM alone into the baseline improves the mIoU from 18.35% to 19.95% on the Vaihingen dataset (with iSAID as the source domain), demonstrating that DDFCM effectively enhances the model's robustness against appearance variations in unseen domains. The performance is further elevated with the subsequent introduction of CFFIM; for instance, the mIoU rises from 24.34% to 25.79% when transferring from DLRSD to Potsdam. This improvement underscores the potent capability of CFFIM in performing cross-modal vision–language alignment within the deep feature space, thereby mitigating semantic shifts in cross-domain scenarios. Finally, the inclusion of the WCAEM module completes the full CDSeg model, which achieves peak per formance.

On the iSAID → Vaihingen task, the mIoU ultimately reaches 22.46%, representing a 4.11% improvement over the baseline. This enhancement suggests that the multidirectional and multiscale feature fusion and decoding strategy of WCAEM effectively refines the feature flow throughout the entire network. To further verify the indispensability of each component, we conducted subtractive ablation experiments by systematically removing specific modules from the complete CDSeg model and observing the resulting performance degradation. As detailed in Table IV, the exclusion of the DDFCM module leads to a significant performance drop when the model encounters datasets with substantial spatial layout discrepancies, such as Vaihingen. Specifically, under the iSAID-trained setting, the mIoU on Vaihingen decreases from 22.46% to 19.82%, underscoring the critical role of DDFCM in bridging the vast domain gap between natural and RS imagery. Moreover, the removal of the CFFIM module results in the most pervasive performance decline. For instance, the mIoU on the Potsdam dataset drops sharply by 1.47%. This reduction highlights that in open-vocabulary tasks, the absence of deep vision–language feature fusion prevents the model from accurately capturing the core semantics of unseen categories, thereby validating the efficacy of CFFIM in promoting generalization across novel scenes and classes. Similarly, the absence of WCAEM leads to a comprehensive regression across all metrics, confirming its importance in final directional and multiscale feature representation. Consequently, we can conclude that the DDFCM, CFFIM, and WCAEM modules in CDSeg are not merely a naive concatenation of components but constitute a logically rigorous and functionally complementary unified architecture. By collectively contributing through domain-invariant feature extraction, cross-modal alignment, and feature enhancement, these modules enable CDSeg to achieve superior generalization capabilities in challenging cross-domain OVSS tasks.

H. Generalization Ability Comparison Experiment Analysis

To rigorously evaluate the generalization robustness of CDSeg in open-vocabulary scenarios, we conduct a comprehensive comparative analysis against the SOTA RSOVSS method, OVRS, alongside several representative closed-set semantic segmentation architectures, including DeepLabv3 [52], SegFormer [53], ST-UNet [54], and FTransUNet [8].

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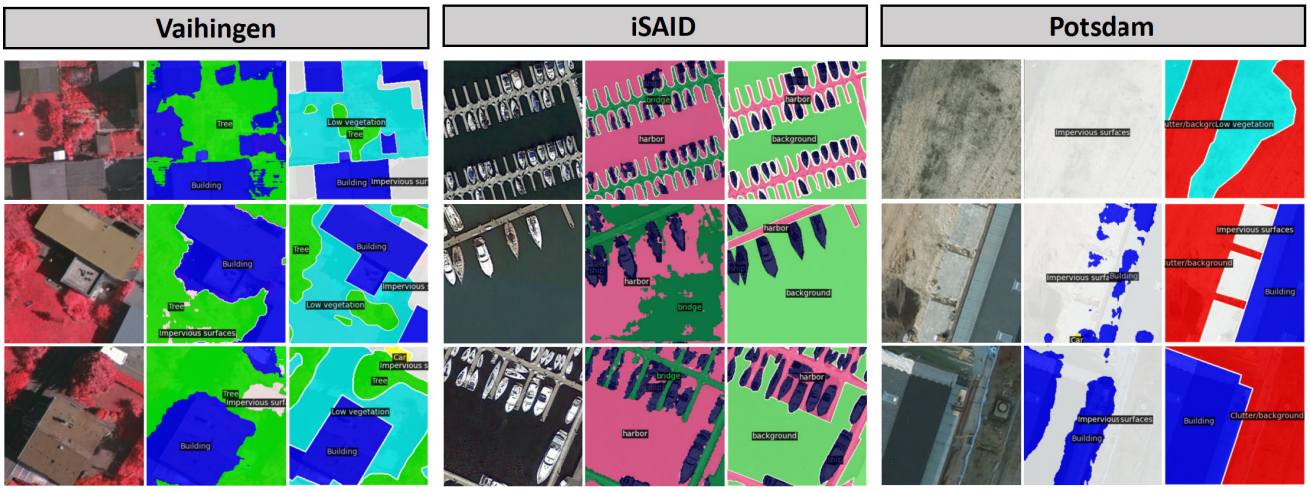


Fig. 8. Training on the DLRSD dataset and test qualitative visualization of inherent failure case analysis on three other datasets. From left to right are the Vaihingen, iSAID, and Potsdam datasets.

Unlike open-vocabulary paradigms, traditional closed-set frameworks are inherently constrained by their fixed-category classification heads, lacking the cross-modal alignment with language models necessary to extract versatile category-specific embeddings. This structural limitation severely restricts their semantic flexibility and diminishes their generalization potential when encountering novel classes. Consequently, as evidenced by the empirical results in Table V, while conventional closed-set models undergo significant performance degradation under cross-domain shifts, both CDSeg and OVRS consistently maintain superior and stable performance. These results underscore the effectiveness of open-vocabulary architectures in capturing domain-invariant features and handling unseen environmental distributions.

I. Failure Cases Analysis

As illustrated in Fig. 8, while CDSeg delivers SOTA segmentation accuracy, its performance is not without deficiencies in cross-dataset testing. To further elucidate this issue, we conducted visual validation on the iSAID, Potsdam, and Vaihingen datasets using the model trained on the DLRSD dataset. As demonstrated in the provided figures, the model fails to effectively decouple "harbor" from "background" features in the iSAID dataset, resulting in substantial interclass confusion. This lack of robustness extends to the Potsdam and Vaihingen datasets, where CDSeg demonstrates diminished discriminative power for "clutter," likely a consequence of the class imbalance (low pixel frequency). Additionally, the intrinsic semantic resemblance between "trees" and "low vegetation" precipitates large-scale misclassification, highlighting a bottleneck in the model's ability to capture fine-grained semantic nuances. The substantial similarity between these two categories presents a formidable challenge; even the supplementary RS-domain knowledge encapsulated within DINOv3 is insufficient to effectively discriminate their intricate semantic relationships and visual commonalities. These challenges represent critical avenues that warrant progressive investigation in future research.

V. Conclusion

In this work, we propose a novel RSOVSS framework, designated as CDSeg. To bridge the substantial feature gap between natural and RS domains and address the limited generalization inherent in open-vocabulary tasks, CDSeg achieves dual-domain knowledge complementarity via a DDFCM, leveraging two prominent vision foundation models: CLIP and DINOv3. Furthermore, we introduce a unified CFFIM to enhance spatial-class generalization capabilities. To address the complexities of varying orientations and scales in RS scenes, CDSeg incorporates a WCAEM during the decoding stage. This module utilizes frequency-domain decomposition to further refine the representation of multiscale and directional features. Extensive experiments demonstrate that CDSeg outperforms various SOTA methods, achieving superior performance. Additionally, comprehensive ablation studies validate the efficacy and indispensability of each constituent module. Future work will focus on lightweight RSOVSS, aiming to strike an optimal balance between model performance and computational efficiency.

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